

Vivi Huang

Computer Engineering, University of Waterloo
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SKILLS

Hardware/HDL: SystemVerilog, RTL/FSM design, pipelining, timing closure, RTL simulation and verification
EDA/Embedded Tools: Vivado, Quartus, cocotb, STM32CubeIDE, logic analyzer, Git, Docker, CI/CD, Linux/Unix
Languages: C/C++, Python, Java, TypeScript, JavaScript, SQL, ArkTS
Libraries/Frameworks: PyTorch, Larq, ONNX Runtime, OpenCV, MediaPipe, FastAPI, React

EDUCATION

•BASc, Computer Engineering

Sept 2023 - Apr 2028

University of Waterloo

GPA: 4.0/4.0

- Dean's Honour List (Fall 2023); Term Distinction in all semesters
- **Courses:** Digital Hardware Systems, Computer Architecture, Digital Circuits, Algorithms and Data Structures

PROJECTS

•Pipelined Matrix-Vector Multiplication Engine — SystemVerilog, Vivado, AMD Kria FPGA

Jul - Aug 2026

- Built a parameterizable **MVM accelerator** (datapath, memory, and control): a **fully pipelined 8-lane dot-product unit** using 8 parallel multipliers feeding a $\log_2 N$ -level **adder reduction tree**, an accumulator, and a two-state **FSM controller** sequencing memory read addresses and datapath valid/first/last control signals
- Met timing at the **650 MHz** target through register-balanced pipelining, and maximized throughput on 512×512 operands by scaling to **128 compute lanes** against the device's **DSP and BRAM** budget under **out-of-context synthesis**
- Wrote per-module **unit testbenches** before top-level integration and resolved datapath/control mismatches from **simulation waveforms**; validated the implemented netlist with **post-implementation functional simulation** and resource-utilization checks

•8-bit Binary Neural Network ASIC — TinyTapeout [Project]

May - Aug 2025

- Designed a **3-layer, 20-neuron** binarized MLP as an **8-bit** digital circuit for tapeout, training the weights offline in **Larq** and hard-mapping them into the RTL so inference reduces to **XNOR and popcount** logic
- Verified the design with **cocotb** testbenches driven from Python, comparing RTL output against the trained model, and gated every change through TinyTapeout's **GitHub Actions CI** hardening and GDS flow

•Gestalt Engine: Real-time Gesture-Audio Interface [GitHub]

Feb 2026 - May 2026

- Implemented a stacked two-layer **LSTM** gesture recognizer twice — **from scratch in NumPy** with a hand-written LSTM cell and backpropagation ($\sim 85\%$ test accuracy) and in **PyTorch** ($\sim 92\%$) — over **21 dynamic gesture classes**, authoring a 500-line pipeline to transform **800k raw IPN-Hand frames** into training-ready arrays
- Served real-time inference at **<15ms per frame** in an **OpenCV** loop, and streamed predictions from a **Vite/React/FastAPI** web app over websockets at **sub-100ms** end-to-end latency

EXPERIENCE

•Software Engineer Intern

Sept 2025 - Apr 2026

Huawei Technologies Canada, Co.

Markham, ON

- Cut model-export runtime on a teammate's **ONNX Runtime** pipeline for a large model by restructuring it around **pthreads** in **C++**, and debugged the file-generation path end to end
- Owned a cycling-route extraction pipeline recovering ordered landmarks via **morphological processing** and route-snap via HuaweiMaps API. Cut runtime from **30s -> 13s** using **AsyncIO** for dual-pipeline architecture.
- Fine-tuned **YOLO26** for in-pipeline map icon detection on a self-curated **2.3k-image** dataset, a class absent from pretraining, reaching **98% mAP@50** with mosaic and flip augmentation
- Built full frontend for multi-person Xiaoyi voice agent gating replies on **voiceprint identity**, and cut p50 response latency from **7-8s to under 3s** by streaming **VAD/LLM/TTS** stages and speaker verification concurrently

•Full Stack Developer

May 2024 - August 2024

Adaptive Pulse

Toronto, ON

- Introduced authentication to a B2B AI SaaS web app. Built a new **OAuth2 SSO** flow over Google Drive and Dropbox APIs so **100+ client companies** could bulk-import documents instead of uploading one file at a time
- Built a **RAG** summarization service over client tax filings and financial reports using **LangChain.js**, OpenAI embeddings, and **pgvector**, cutting document review time by $\sim 30\%$ in internal benchmarking